Suyash Gaikwad

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SUMMARY

Photonics and Optoelectronics M.S. student at National Taiwan University (thesis: semiconductor-based SERS) with a computer engineering background, industry experience in automation and 24/7 reliability (Red Hat, Vinsys) and published research. Seeking a 2027 summer internship in semiconductor process, metrology or yield engineering.

EDUCATION

M.S., Photonics and Optoelectronics Sep 2026 – Jun 2028 (expected)
National Taiwan University (NTU), Graduate Institute of Photonics and Optoelectronics, Taipei, Taiwan

- Thesis (in progress): semiconductor-based surface-enhanced Raman spectroscopy (SERS) substrates.
- Coursework: Physics of Semiconductor Devices; Light-Emitting Semiconductor Physics; Light Scattering Simulation.
- Ministry of Education (MOE) Taiwan Scholarship recipient.

B.Tech., Computer Engineering Dec 2021 – May 2024
Vishwakarma Institute of Technology (Savitribai Phule Pune University), Pune, India

- CGPA 8.63/10. Coursework: Machine Learning, Data Science, Artificial Intelligence, Algorithms.

Diploma, Computer Engineering – Marathwada Mitra Mandal's Polytechnic, Pune (95.49%) Nov 2017 – Aug 2021

RESEARCH EXPERIENCE

Research Intern, Taiwan Experience Education Program (TEEP) Jun 2023 – Nov 2023
College of EECS, National Tsing Hua University (Advisor: Prof. Cheng-Hsin Hsu), Hsinchu, Taiwan

- Designed and implemented a digital-twin model of a wind turbine in Python, simulating performance under varying environmental conditions with AWS IoT Core, AWS IoT SiteWise and Jupyter.
- Built real-time sensor-data processing and condition-monitoring dashboards; predictive analysis showed up to 20% simulated efficiency improvement and 30% faster fault detection.
- Integrated IoT sensor data with cloud visualization in a multidisciplinary team; published at ACM (2024).

PROFESSIONAL EXPERIENCE

OpenShift Platform Engineer Mar 2025 – Dec 2025
Vinsys, Pune, India

- Supported the migration to Red Hat OpenShift, managing ArgoCD and Tekton CI/CD pipelines (Red Hat Fuse).
- Ran proactive health checks and backups to keep distributed production systems at 24/7 uptime.
- Built Excel-based automation tools for operational monitoring, reducing manual intervention by 30%.

OpenShift Platform Engineer Intern Jun 2024 – Aug 2024
Red Hat, Pune, India

- Monitored and supported OpenShift clusters with DevOps teams, improving visibility into containerized workloads.
- Coordinated ServiceNow ticket triage and assisted with infrastructure audits and compliance checks.

PUBLICATIONS

- C.-C. Lai, A. Sarkar, P. A. Dinh, S. Gaikwad, C.-H. Hsu. "Urban Digital-Twin Planning for Sustainable Smart Cities: System Architecture, Preliminary Experiments, and Open Challenges." Proc. 2nd International Workshop on Middleware for Digital Twins, ACM, Dec 2024, pp. 7-12. doi:10.1145/3702636.3703443
- S. Gaikwad et al. "Fingerprint Based Door Lock Using Arduino." IRJMETS, vol. 3, no. 6, Jun 2021. Software copyright SW-15040/2021 (India).

PROJECTS AND OUTREACH

- Teaching on YouTube: 10 Linux (RHEL) tutorials and a self-hosted CI/CD homelab build (Jenkins, Docker, Tailscale).
- Smart street light: energy-saving street-light system; Best Project Award, AVANT, MM Polytechnic (2019).
- SAARTHI volunteer (Jun 2022 – Jul 2022): taught new technology to full-time homemakers.

SKILLS

Semiconductors and optics: semiconductor devices, light-emitting semiconductors, light-scattering simulation, SERS
Programming and data: Python, C/C++, Java, SQL, TensorFlow, PyTorch, scikit-learn, Jupyter, data visualization
Automation and systems: Linux (RHEL), Docker, Kubernetes, OpenShift, ArgoCD, Tekton, Jenkins, Git, Grafana, AWS IoT
Hardware and IoT: Arduino, Raspberry Pi, ESP32, sensor integration, Node-RED, ThingSpeak
Languages: English (professional), Mandarin Chinese (beginner), Hindi (native), Marathi (native)